

subscriber trunk dialling (STD) calls and data services. Signals from the towers are sent to the core network, and the latter is responsible for the execution of the final service. This can be any kind of service, even a simple Internet connection. So the core network acts as a gateway between the radio access network and the final services (Figure 1).

Traditional RAN (radio access network)

In a traditional RAN, the base station consists of mainly two units. The first is the radio unit (RU) or the signal transmitting unit and the second is the baseband unit (BBU), which is responsible for calculating the signals. The RU receives the signals and sends them to the BBU. These are then sent to the core network. A single vendor provides all the units and the connection between them is deployed only by that vendor. This is a very tightly coupled system, which makes it hard to change even a single component. The software is very tightly integrated inside the hardware, which presents a challenge to change it, making it difficult to upgrade the infrastructure. This increases the dependency on a single vendor, and one has to shift to a different vendor to change the entire hardware and the software, which is never convenient.

So, what is Open RAN?

Open RAN is a set of specifications for desegregating software and hardware. It allows hardware and software from different vendors to work together. Servers can now independently implement virtualisation due to this desegregation. Open RAN even makes cloud solutions possible. It lets one virtualised baseband unit support multiple radios instead of needing a proprietary or a physical unit at every cell site. This makes the infrastructure flexible and cost effective.

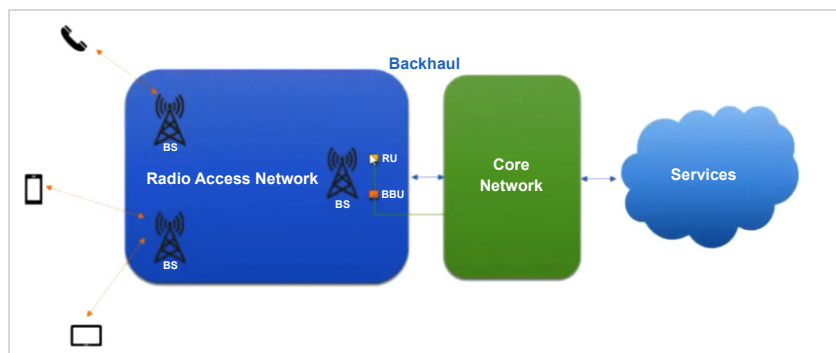


Figure 1: Mobile network

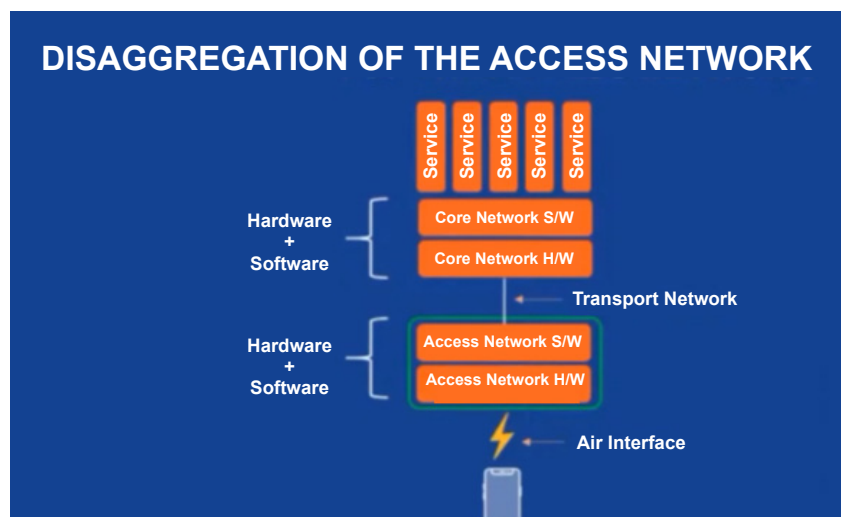


Figure 2: Typical mobile architecture

But why should one go for Open RAN (O-RAN)?

- **Hardware and software disaggregation:** There are no direct restrictions on the hardware and the software.
- **Intelligence and automation:** It makes management and orchestration more feasible since everything is quite open and there are standard rules to do things. It uses artificial intelligence and machine learning capabilities.
- **Open internal RAN interfaces:** The O-RAN lower layer basically splits a single unit into multiple units, mobilises them and makes it a multi-vendor solution.
- **Rapid innovation:** O-RAN has improved the innovation pace

significantly due to the increased competition among vendors themselves. New companies are now coming up with their own components which tend to work a lot better.

Architecture of a typical mobile network

As can be seen in Figure 2, there is a core network and an access network, which is connected by the transport network (also known as the backhaul network). Then there is an air interface that connects a user with the service provider's network. There are various other services that run on top of the core network. The core network and the access network contain both the hardware and software, which can be seen on the left in Figure 2.